

Bharadwaja O

Software Engineer -1

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Professional Summary

Results-driven Software Engineer with **4+ years of experience** and track record in developing high-performance applications and scalable solutions across e-commerce, financial, and technology industries. Expertise in **Core Java, Spring Boot, Microservices**, and **full-stack development**, with strong skills in **front-end** frameworks like **React** and **backend** technologies. Adept at integrating complex systems, optimizing database performance, and delivering secure, efficient, and maintainable code.

Education

Master of Science in Computer Science

University of Wisconsin – Milwaukee, Wisconsin | sept. 2021 – Dec. 2022

Skills

- **Programming Languages:** Java, Python, JavaScript, TypeScript, SQL, C++
- **Frameworks & Tools:** Spring Boot, Spring MVC, Spring Security, Hibernate, J2EE, React.js, Redux, Node.js
- **Web Technologies:** REST APIs, SOAP, GraphQL, XML, JSON, AJAX, JSP, Servlets, WebSockets, gRPC
- **Databases:** PostgreSQL, Oracle DB, MySQL, SQL Server, MongoDB, Cassandra, DynamoDB
- **Cloud & DevOps:** AWS (EC2, S3, RDS, Lambda, CloudWatch, ELB), Azure, Terraform, Docker, Kubernetes, Jenkins, CI/CD Pipelines
- **Testing:** JUnit, Mockito, Cypress, Playwright, Selenium, JMeter, LoadRunner
- **Messaging & Event Streaming:** Apache Kafka, RabbitMQ, JMS, Azure Event Hubs
- **Monitoring & Logging:** Log4j, ELK Stack (Elasticsearch, Logstash, Kibana), Splunk
- **Version Control & Collaboration:** Git, GitHub, GitLab, Bitbucket, Jira, Confluence

Professional Experience

Software Engineer-1

Microsoft, Seattle | Jan 2024 – Present

- Architected low-latency AI orchestration layers using **Java (Spring Boot)**, **gRPC**, and **WebSockets**, ensuring near-instantaneous **Copilot response times (<50ms)** for real-time user interactions.
- Engineered a dynamic AI prompt pipeline by integrating **LLM fine-tuning models with Azure OpenAI Service**, optimizing **Copilot's contextual awareness** based on browsing patterns.
- Implemented a federated learning framework to personalize **Copilot's AI recommendations** while ensuring **privacy-first AI inference** using **differential privacy techniques** and **secure multi-party computation (MPC)**.
- Optimized microservices scaling strategies via **Kubernetes (AKS)**, **Knative**, and **Azure Service Fabric**, achieving **40% cost reduction** in dynamic workload management.
- Developed a **distributed caching layer** using **Azure Redis Cache** and **CQRS pattern**, improving the **Copilot AI suggestion retrieval time by 60%**.
- Designed an event-driven AI pipeline leveraging **Apache Kafka**, **Azure Event Hubs**, and **Debezium**, enabling real-time updates for **Copilot's contextual understanding** across browsing sessions.
- Implemented Just-in-Time AI Model Deployment with **KServe** and **ONNX Runtime**, ensuring **seamless LLM upgrades** without user disruption.
- Integrated Edge's AI-assisted summarization with **Azure Cognitive Search** and **Vector Databases (Weaviate, FAISS)** to enhance **Copilot's contextual responses** for complex queries.
- Developed a predictive **auto-scaling strategy** using **Azure Machine Learning** and **Prometheus**, dynamically adjusting AI service instances based on traffic patterns and reducing **cold-start latency by 35%**.

Software Engineer

Northwestern Mutual, Milwaukee | Dec 2022 – Jan 2024

- Developed **RESTful microservice** using **Spring Boot, Spring Data JPA, and Spring Cloud** to build scalable financial internal applications.
- Implemented **Spring MVC Controllers** and configured **Spring Security** with **OAuth2** for secure authentication and authorization.
- Built dynamic UI components using **Angular 5, Node.js, HTML5, CSS, and JavaScript**, enhancing user experience.
- Utilized **SQL Server** for data persistence, optimizing queries and integrating backend services with the frontend.
- Designed and configured **Jenkins CI/CD pipelines** for automated builds and deployments.
- Worked with **Git and Bitbucket** for version control and collaborated in an **Agile development** environment.
- Used **Docker and Nginx** for containerized deployments and load balancing.
- Deployed infrastructure on **AWS (EC2, RDS, S3, ELB, CloudWatch)** for high availability and scalability.

Associate Software Engineer

Dxc technology, Hyderabad, India | May 2020 – Aug 2021

- Developed **full-stack web applications** using Java (**Spring Boot, Spring MVC, Spring Security**) for backend and **ReactJS, Node.js, and AJAX** for frontend, ensuring seamless user interactions and real-time data updates.
- Built and optimized **RESTful APIs** with **Spring Data JPA, Hibernate, and JAX-RS**, enabling efficient communication between client and server layers.
- Configured and optimized databases like **Oracle, MySQL, PostgreSQL, MongoDB, Cassandra, and DynamoDB** for scalable and high-performance data storage.
- Utilized **JDBC, JNDI, and JMS** to handle transactions, manage resources, and ensure reliable messaging in distributed systems.
- Created **unit tests** with **JUnit and Mockito**, improving code quality and reducing production issues.
- Managed **CI/CD pipelines** using **Maven, Gradle, and Jenkins**, streamlining the software development lifecycle.
- Deployed applications on **AWS (EC2, S3, Lambda)** and integrated **MongoDB & Cassandra** for large-scale data processing.
- Designed **interactive UI components** using **JSP, Servlets, HTML5, CSS3, JavaScript, TypeScript, Bootstrap, and jQuery**, enhancing user experience.

Projects

Knowledge-Enhanced Chatbot Interface

- Designed and implemented a chatbot using OpenAI's LLM models integrated with AWS services, improving query accuracy and operational efficiency.

Certifications

- Microsoft Certified: DevOps Engineer Expert
- Oracle Certified Professional: Java SE 11 Developer
- Certified Kubernetes Administrator

Highlights

- Increased transaction success rates by 98% through optimized microservices.
- Reduced operational time by 2 hours/day with AI-driven solutions.